

Vivek Shanbhag

SYSTEMS DEVELOPMENT ENGINEER II | SOFTWARE, CLOUD & AGENTIC AI

United States | vivekduq@gmail.com | [linkedin.com/in/vivshanbhag](https://www.linkedin.com/in/vivshanbhag) | vivshanbhag.github.io

PROFESSIONAL SUMMARY

Software and systems engineer with 11+ years of experience building software, automation, cloud, data, and machine-learning solutions across operational and customer-facing environments. At AWS, work focuses on system design, operational reliability, automation, and AI-enabled workflows. Experienced in translating complex workflows into reliable, reviewable systems; M.S. in Analytics and Information Management.

TECHNICAL SKILLS

Software: Python, Rust, TypeScript, Java, JavaScript, REST APIs, SQL, backend and full-stack development, system design, unit testing, automation

Cloud and Systems: AWS (EMR, Glue, Lambda, S3, EC2, DynamoDB, IAM, VPC), AWS CDK, distributed systems, cloud architecture, infrastructure as code, observability, reliability engineering

AI and Data: Agentic AI, LLM tool use, workflow orchestration, Amazon Bedrock, SageMaker, machine learning, NLP, XGBoost, TensorFlow, scikit-learn, Kafka, Spark, Hadoop, Databricks, Pandas, Tableau, Power BI

PROFESSIONAL EXPERIENCE

Systems Development Engineer II

Mar 2024 - Present

Amazon Web Services | Seattle, WA

- Build and improve reliable systems, automation, and internal engineering tools for complex operational workflows.
- Apply system design, AI-enabled workflows, operational visibility, and on-call reliability practices; employer work is described at a high level to respect confidentiality.

Forward Deployed Engineer - EMR

Aug 2022 - Mar 2024

Amazon Web Services | Boston, MA

- Supported AWS customers with cloud architecture, migration, and troubleshooting for Amazon EMR and distributed data workloads.
- Developed engineering automation and published technical guidance; served as a Quality Champion for the NA-East region.

Data Engineering Intern

May 2022 - Aug 2022

Highmark Health | Pittsburgh, PA

- Collaborated on a data-pipeline migration from Informatica BDM to Google Cloud using Kafka and Databricks.
- Built an ARIMA model for irregular insurance-claims patterns, reducing database anomalies by 8% and improving KPIs by 26%; created Tableau views for pricing and sales decisions.

Data Scientist Intern

Apr 2022 - May 2022

Carmeuse Americas | Pittsburgh, PA

- Developed machine-learning models to predict usable lime production and automated weekly, site-level reporting for operations managers.

PROFESSIONAL EXPERIENCE - CONTINUED

Data Science Intern

Jul 2021 - Mar 2022

BioMotivate | Pittsburgh, PA

- Applied statistical and machine-learning methods to healthcare and wearable-sensor data.
- Built ETL pipelines and an XGBoost classifier with 85% accuracy for a treatment-dropout prediction task; created reusable resampling code and KPI dashboards for clinical staff.

Graduate Research Assistant

May 2021 - Mar 2022

Duquesne University | Pittsburgh, PA

- Supported faculty research in machine learning and data engineering, contributed to data-collection design, and built Python models for sports analytics.

Information Technology Analyst / Team Lead

Dec 2016 - Apr 2021

Tata Consultancy Services | Mumbai, India

- Led a seven-person software development and consulting team on the TCS iON Energy project.
- Designed a ticket-escalation module that saved more than 200 hours annually and contributed to a 6.4% revenue increase.
- Built executive dashboards and energy analytics modules; development guidelines and training contributed to a 15% productivity increase. Received the Best Process Improvement Award in April 2020.

Product Engineer

Apr 2014 - Dec 2016

Intellect Design Arena | Mumbai, India

- Automated multilingual report generation, reducing a client's manual reporting workload by 80% (about 10 hours per week).
- Developed a Java Spring and REST currency-exchange calculator; supported requirements analysis, unit testing, technical documentation, and WebLogic upgrades.

OPEN-SOURCE PROJECT

Incident Triage Agent | Human-in-the-loop agentic incident-response reference app. Uses bounded, read-only diagnostics to produce an evidence-based incident brief for operator review. Includes high-level and low-level design docs.

github.com/vivshanbhag/incident-triage-agent

EDUCATION

Master of Science in Analytics & Information Management | Duquesne University | 2022 | Summa Cum Laude, GPA 4.0 / 4.0

Master of Business Administration, Finance | Welingkar Institute of Management | 2019 | Distinction; ranked first in Finance batch

Bachelor of Engineering in Computer Engineering | University of Pune | 2013 | First Class

AWARDS AND CERTIFICATIONS

AWSome Builder Award, AWS GSSO Innovation Hackathon winner (2025); AWSome Builder Awards (June and August 2024); Python Certification, HackerRank (2024); Intro to Python, O'Reilly (2024); HTML5, JavaScript & CSS3 Specialist, Microsoft (2013).